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Bladed rotor inspection, analysis and repair systems and methods

Abstract

A method can comprise: determining, via a processor, whether serviceable limits are met for an inspected bladed rotor; generating, via the processor, a three-dimensional model including a repair blend profile for a defect in response to determining the inspected bladed rotor does not meet the serviceable limits; performing, via the processor, a first set of simulations on the three-dimensional model; determining, via the processor, whether a first set of results of the first set of simulations meet an experience based criteria; performing, via the processor, a second set of simulations on the three-dimensional model in response to a first result in the first set of results not meeting a first objective parameter in the experience based criteria; and determining, via the processor, whether a second set of results and the first set of results meet a deterministic criteria.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a non-provisional of, and claims priority to, and the benefit of U.S. Provisional Application No. 63/327,748, entitled “BLADED ROTOR INSPECTION, ANALYSIS AND REPAIR SYSTEMS AND METHODS,” filed on Apr. 5, 2022, which is hereby incorporated by reference in its entirety for all purposes.

FIELD

(1) The present disclosure relates to repair analysis methods and systems, and more particularly, the repair analysis systems and methods for bladed rotors of gas turbine engines.

BACKGROUND

(2) Gas turbine engines (such as those used in electrical power generation or used in modern aircraft) typically include a compressor, a combustor section, and a turbine. The compressor and the turbine typically include a series of alternating rotors and stators. A rotor generally comprises a rotor disk and a plurality of blades. The rotor may be an integrally bladed rotor (“IBR”) or a mechanically bladed rotor.

(3) The rotor disk and blades in the IBR are one piece (i.e., monolithic, or nearly monolithic) with the blades spaced around the circumference of the rotor disk. Conventional IBRs may be formed using a variety of technical methods including integral casting, machining from a solid billet, or by welding or bonding the blades to the rotor disk.

(4) Repair methods for IBRs may be limited to maintaining the IBR within tolerances of a product definition for a design of the IBR, a volume or thickness of material addition, and/or a volume or thickness of material removal. Additionally, repair analysis methods for IBRs may fail to consider loads and/or boundary conditions encountered during engine operation included in the aerodynamic, structural, or other functional assessment of the IBR. In this regard, a damaged IBR may be scrapped because it is believed to unrepairable within the geometric assessment constraints (e.g., tolerances, material removal, material addition, etc.) without consideration of the functional assessment constraints (e.g., aerodynamic stability, structural durability, etc.). By being limited in this manner, less expensive repair options and/or more optimal repairs (e.g., in terms of aerodynamic or structural capability, etc.) that could otherwise be acceptable may be avoided.

SUMMARY

(5) A method is disclosed herein. The method can comprise: determining, via a processor, whether serviceable limits are met for an inspected bladed rotor; automatically generating, via the processor, a three-dimensional model including a repair blend profile for a defect in response to determining

the inspected bladed rotor does not meet the serviceable limits; automatically performing, via the processor, a first set of simulations on the three-dimensional model; automatically determining, via the processor, whether a first set of results of the first set of simulations meet an experience based criteria; automatically performing, via the processor, a second set of simulations on the three-dimensional model in response to a first result in the first set of results not meeting a first objective parameter in the experience based criteria; and automatically determining, via the processor, whether a second set of results and the first set of results meet a deterministic criteria.

(6) In various embodiments, the method further comprises repairing the inspected bladed rotor with the repair blend profile in response to determining the first set of results and the second set of results meet the deterministic criteria. In various embodiments, the method further comprises modifying the experience-based criteria in response to determining the first set of results and the second set of results meet the deterministic criteria.

(7) In various embodiments, the experience-based criteria includes a modal assurance criteria (MAC), a resonant frequency, an aerodynamic efficiency, and a stall margin.

(8) In various embodiments, the method further comprises receiving, via the processor, a digital point cloud of the inspected bladed rotor. In various embodiments, the method further comprises automatically converting, via the processor, the digital point cloud to the three-dimensional model with the repair blend profile for the defect in response to determining the inspected bladed rotor does not meet the serviceable limits.

(9) In various embodiments, the serviceable limits include at least one of a threshold defect depth, a threshold defect length, a threshold defect aspect ratio, a threshold number of defects per blade, a threshold number of defects per bladed rotor.

(10) A method is disclosed herein. The method can comprise: inspecting, via an inspection system, an inspected bladed rotor; analyzing, via an analysis system, the inspected bladed rotor, the analyzing including: determining whether serviceable limits are met for the inspected bladed rotor; determining whether experience based criteria is met for a repaired bladed rotor in response to the inspected bladed rotor not meeting the serviceable limits, the repaired bladed rotor including a repair blend profile for repairing a defect; and determining whether deterministic criteria is met for the repaired bladed rotor in response to the repaired bladed rotor not meeting the serviceable limits; and repairing, via a repair system, the inspected bladed rotor to form the repaired bladed rotor in response to determining the inspected bladed rotor does not meet the serviceable limits and determining one of experience based criteria for the repaired bladed rotor being met or the deterministic criteria for the repaired bladed being met.

(11) In various embodiments, the inspecting is performed with a structured scanner to form a point cloud of the inspected bladed rotor. In various embodiments, the analyzing includes generating a repaired bladed rotor digital model of the repaired bladed rotor based on the point cloud and determining an initial blend shape for the repair blend profile. In various embodiments, the analyzing further comprises performing a set of simulations on the repaired bladed rotor digital model. In various embodiments, the analyzing further comprises comparing a set of results from the set of simulations to the experience based criteria. In various embodiments, the repairing comprises performing a repair on the inspected bladed rotor to generate the repair blend profile in response to the experience based criteria being met by the set of results from the set of simulations. In various embodiments, the experience based criteria includes a modal assurance criteria (MAC), a resonant frequency, an aerodynamic efficiency, and a stall margin.

(12) A system is disclosed herein. The system can comprise: an inspection system configured to generate a point cloud of at least a portion of an inspected bladed rotor; and an analysis system in electronic communication with the inspection system, the analysis system comprising a tangible, non-transitory computer-readable storage medium having instructions stored thereon that, in response to execution by a processor, cause the processor to perform operations comprising: receive, via the processor, the point cloud; determine, via the processor and based on the point

cloud, whether serviceable limits are met for the inspected bladed rotor; generate, via the processor, a three-dimensional model with a repair blend profile for a defect in the inspected bladed rotor in response to determining the inspected bladed rotor does not meet the serviceable limits; perform, via the processor, a set of simulations of the three-dimensional model; and determine, via the processor, whether an experience based criteria is met by a first set of results of the set of simulations.

(13) In various embodiments, the system further comprises the inspection system comprises a scanner.

(14) In various embodiments, the experience based criteria includes a modal assurance criteria (MAC), a resonant frequency, an aerodynamic efficiency, and a stall margin.

(15) In various embodiments, the operations further comprise, performing additional simulations, via the processor, in response to determining the first set of results do not meet the experience based criteria.

(16) In various embodiments, the operations further comprise, sending repair instructions to the inspection system in response to determining the first set of results meet the experience based criteria. The inspection system can comprise a computer numerical control (CNC) machine, the CNC machine configured to perform the repair instructions.

(17) The foregoing features and elements may be combined in any combination, without exclusivity, unless expressly indicated herein otherwise. These features and elements as well as the operation of the disclosed embodiments will become more apparent in light of the following description and accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The subject matter of the present disclosure is particularly pointed out and distinctly claimed in the concluding portion of the specification. A more complete understanding of the present disclosure, however, may best be obtained by referring to the following detailed description and claims in connection with the following drawings. While the drawings illustrate various embodiments employing the principles described herein, the drawings do not limit the scope of the claims.

(2) FIG. 1A illustrates a cross-sectional view of a gas-turbine engine, in accordance with various embodiments;

(3) FIG. 1B illustrates a cross-sectional view of a high pressure compressor, in accordance with various embodiments;

(4) FIG. 2 illustrates a front view of a bladed rotor, in accordance with various embodiment;

(5) FIG. 3 illustrates a portion of an inspected bladed rotor, in accordance with various embodiments;

(6) FIG. 4 illustrates a portion of the inspected bladed rotor during repair, in accordance with various embodiments;

(7) FIG. 5A illustrates a portion of a repaired bladed rotor, in accordance with various embodiments;

(8) FIG. 5B illustrates a repair blend profile, in accordance with various embodiments;

(9) FIG. 5C illustrates a repair blend profile, in accordance with various embodiments;

(10) FIG. 5D illustrates a repair blend profile, in accordance with various embodiments;

(11) FIG. 6A illustrates an inspection, analysis, and repair process for a bladed rotor, in accordance with various embodiments;

(12) FIG. 6B illustrates an inspection, analysis, and repair process for a bladed rotor, in accordance with various embodiments;

- (13) FIG. 6C illustrates an analysis during the inspection, analysis, and repair process for a bladed rotor from FIGS. 6A-B, in accordance with various embodiments;
- (14) FIG. 6D illustrates an inspection, analysis, and repair process for a bladed rotor, in accordance with various embodiments;
- (15) FIG. 6E illustrates an inspection, analysis, and repair process for a bladed rotor, in accordance with various embodiments;
- (16) FIG. 7, illustrates a system to perform the inspection, analysis, and repair process of FIGS. 6A and 6B, in accordance with various embodiments;
- (17) FIG. 8 illustrates an inspection and/or repair system, in accordance with various embodiments;
- (18) FIG. 9 illustrates a schematic view of a control system for an inspection system, in accordance with various embodiments;
- (19) FIG. 10 illustrates a schematic view analysis system, in accordance with various embodiments; and
- (20) FIG. 11 illustrates a process for on-demand repair dispositions, in accordance with various embodiments.

DETAILED DESCRIPTION

(21) The following detailed description of various embodiments herein refers to the accompanying drawings, which show various embodiments by way of illustration. While these various embodiments are described in sufficient detail to enable those skilled in the art to practice the disclosure, it should be understood that other embodiments may be realized and that changes may be made without departing from the scope of the disclosure. Thus, the detailed description herein is presented for purposes of illustration only and not of limitation. Furthermore, any reference to singular includes plural embodiments, and any reference to more than one component or step may include a singular embodiment or step. Also, any reference to attached, fixed, connected, or the like may include permanent, removable, temporary, partial, full or any other possible attachment option. Additionally, any reference to without contact (or similar phrases) may also include reduced contact or minimal contact. It should also be understood that unless specifically stated otherwise, references to “a,” “an,” or “the” may include one or more than one and that reference to an item in the singular may also include the item in the plural. Further, all ranges may include upper and lower values and all ranges and ratio limits disclosed herein may be combined.

(22) As used herein, “aft” refers to the direction associated with the tail (e.g., the back end) of an aircraft, or generally, to the direction of exhaust of the gas turbine. As used herein, “forward” refers to the direction associated with the nose (e.g., the front end) of an aircraft, or generally, to the direction of flight or motion.

(23) With reference to FIG. 1A, a gas turbine engine 20 is shown according to various embodiments. Gas turbine engine 20 may be a two-spool turboprop that generally incorporates a fan section 22, a compressor section 24, a combustor section 26, and a turbine section 28. In operation, fan section 22 can drive air along a path of bypass airflow B while compressor section 24 can drive air along a core flow path C for compression and communication into combustor section 26 then expansion through turbine section 28. Although depicted as a turboprop gas turbine engine 20 herein, it should be understood that the concepts described herein are not limited to use with turboprops as the teachings may be applied to other types of turbine engines including three-spool architectures, single spool architecture or the like.

(24) Gas turbine engine 20 may generally comprise a low speed spool 30 and a high speed spool 32 mounted for rotation about an engine central longitudinal axis A-A' relative to an engine static structure 36 or engine case via several bearing systems 38, 38-1, etc. Engine central longitudinal axis A-A' is oriented in the Z direction on the provided X-Y-Z axes. It should be understood that various bearing systems 38 at various locations may alternatively or additionally be provided, including for example, bearing system 38, bearing system 38-1, etc.

(25) Low speed spool 30 may generally comprise an inner shaft 40 that interconnects a fan 42, a

low pressure compressor **44** and a low pressure turbine **46**. Inner shaft **40** may be connected to fan **42** through a geared architecture **48** that can drive fan **42** at a lower speed than low speed spool **30**. Geared architecture **48** may comprise a gear assembly **60** enclosed within a gear housing **62**. Gear assembly **60** couples' inner shaft **40** to a rotating fan structure. High speed spool **32** may comprise an outer shaft **50** that interconnects a high pressure compressor **52** and high pressure turbine **54**. A combustor **56** may be located between high pressure compressor **52** and high pressure turbine **54**. A mid-turbine frame **57** of engine static structure **36** may be located generally between high pressure turbine **54** and low pressure turbine **46**. Mid-turbine frame **57** may support one or more bearing systems **38** in turbine section **28**. Inner shaft **40** and outer shaft **50** may be concentric and rotate via bearing systems **38** about the engine central longitudinal axis A-A', which is collinear with their longitudinal axes. As used herein, a "high pressure" compressor or turbine experiences a higher pressure than a corresponding "low pressure" compressor or turbine.

(26) The core airflow may be compressed by low pressure compressor **44** then high pressure compressor **52**, mixed and burned with fuel in combustor **56**, then expanded over high pressure turbine **54** and low pressure turbine **46**. Turbines **46**, **54** rotationally drive the respective low speed spool **30** and high speed spool **32** in response to the expansion.

(27) In various embodiments, and with reference to FIG. **1B**, high pressure compressor **52** of the compressor section **24** of gas turbine engine **20** is provided. The high pressure compressor **52** includes a plurality of blade stages **101** (i.e., rotor stages) and a plurality of vane stages **105** (i.e., stator stages). The blade stages **101** may each include an integrally bladed rotor ("IBR") **100**, such that the blades **103** and rotor disks **102** are formed from a single integral component (i.e., a monolithic component formed of a single piece). Although described herein with respect to an IBR **100**, the present disclosure is not limited in this regard. For example, the inspection, analysis, and repair systems disclosed herein can be utilized with bladed rotors formed of separate blades **103** and rotor disks **102** and still be within the scope of this disclosure.

(28) The blades **103** extend radially outward from the rotor disk **102**. The gas turbine engine **20** may further include an exit guide vane stage **106** that defines the aft end of the high pressure compressor **52**. Although illustrated with respect to high pressure compressor **52**, the present disclosure is not limited in this regard. For example, the low pressure compressor **44** may include a plurality of blade stages **101** and vane stages **105**, each blade stage in the plurality of blade stages **101** including the IBR **100** and still be within the scope of this disclosure. In various embodiments, the plurality of blade stages **101** form a stack of IBRs **110**, which define, at least partially, a rotor module **111** of the high pressure compressor **52** of the gas turbine engine **20**.

(29) Referring now to FIG. **2**, a front view of an IBR **100** is illustrated, in accordance with various embodiments. The IBR **100** includes a rotor disk **102** and a plurality of blades **103** extending radially outward from the rotor disk **102**.

(30) When debris is ingested into the gas turbine engine **20**, the debris can pass into the primary flowpath. Due to the rotation of the blades **103** in the primary flowpath, the debris can contact one or more of the blades **103**. This contact can cause damage or wear to a blade **103**, or a set of the blades **103**. Disclosed herein is are systems and methods for inspection, analysis, and repair of an IBR **100** and for returning an IBR **100** back to service after use. The systems and methods disclosed herein facilitate faster dispositions the more the process is utilized. In this regard, the systems and methods disclosed herein provide a feedback system for continuous improvement of the repair process, in accordance with various embodiments.

(31) With combined reference to FIGS. **2** and **3**, a damaged portion **130** from FIG. **2** of an IBR **100** including a substantial number of defects **140** (e.g., damage, wear, etc.) resulting from use of the IBR **100** in the gas turbine engine **20** from FIG. **1A** over time. The size and shape of the defects **140** illustrated in FIG. **3** are exaggerated for illustrative effect. Further, the defects **140** can extend to all of the blades **103** of the IBR **100**, the rotor disk **102**, a set of the blades **103** of the IBR **100**, a single blade in the blades **103**, or the like.

(32) In order to repair the defect **140**, a blending operation can be performed on the IBR **100**. A blending operation uses a material removal process, such as milling or computer numerical control (CNC) machining, to remove the damaged portion of the IBR **100** and smooth the resulting voids such that the IBR **100** can be re-introduced into service for further use.

(33) Referring now to FIG. **4**, the damaged portion **130** with an additional blending mask **150** applied to each of the locations of the defects **140** is illustrated, in accordance with various embodiments. As with the locations of the defects **140**, the blending masks **150** are highly exaggerated in scale for explanatory effect. The blending masks **150** can be physical masking applied to the IBR **100**, or shaded colors on computer simulations of the IBR **100**. In either case, the blending masks **150** indicate what portions of the material of the blade **103** and/or the rotor disk **102** should be removed in order for the IBR **100** to be suitable for utilization in the gas turbine engine **20** after blending. The blending mask **150** ensures that accurate and consistent blends are made in operations that require or utilize manual removal of material.

(34) Once the blending mask **150** has been applied, material is removed using the generally manual material removal operation resulting in a repaired blade portion **172** of a repaired IBR **170** including a plurality of repair blend profiles **160**, as is illustrated in FIG. **5A**.

(35) In various embodiments, each repair blend profile in the plurality of repair blend profiles **160** is based, at least partially, on a defect shape of a respective defect **140** from FIG. **3**. In various embodiments, as described further herein, the systems and methods disclosed herein facilitate a greater number of potential blend options for a respective defect **140** from FIG. **3**. For example, if a repair blend was too large and fell out of experience based functional criteria, an IBR **100** could be scrapped as opposed to being blended as described further herein and placed back into service.

(36) In various embodiments, explicit instructions are derived from the automated process and supplied to a computer numerical controlled (CNC) machine. In this instance, blending masks may not be utilized as the manual blending operation is replaced by the machine automated process. Blends of IBR **100** can be by either manual or automated processes, or a combination of those processes. In wholly automated processes, the creation of the mask can be omitted.

(37) In addition to removing the locations of the defects **140**, it is beneficial to remove material deeper than the observed damage in order to ensure that all damage is removed and to prevent the propagation of new damage. In various embodiments, a blend aspect ratio (e.g., a length-depth ratio) is maintained in order to ensure that there is a smooth and gradual transition from the edge of the undamaged blade **103** to the bottom of the deepest portion of the blend, and then back to the undamaged surface of the blade **103** on the other side of the blend.

(38) Although FIGS. **2-5A** illustrated blending solutions applied to the IBR **100** in locations including the rotor disk **102**, the blade **103**, an edge **121** (e.g., a leading edge or a trailing edge), a tip **122**, an airfoil surface **123** (e.g., a pressure surface or a suction surface), it can be appreciated that a repair of each location shown on the same blade **103** is unlikely. The various locations are shown for illustrative purposes of potential repair locations, in accordance with various embodiments. The illustrated defects **140**, blending masks **150**, and repair blend profiles **160** on the IBR **100** are exemplary of the limited applications and not of every repair made according to the description herein.

(39) In order to assist with the blending process and, facilitate efficient determination of acceptability of an IBR **100** to return into service, a semi-automated (or fully automated) system is utilized to inspect, analyze, and repair an IBR **100**, in accordance with various embodiments.

(40) In various embodiments, repair blend profiles **160** can include a scallop shape, a tear drop shape, a material reduction along an edge (i.e., chord reduction), or the like. The present disclosure is not limited in this regard.

(41) For example, with reference now to FIGS. **5B**, and **5C**, a repair blend profile **162** is illustrated. In various embodiments, the repair blend profile comprises a convex shape and defines a recess in an outer surface of the repaired IBR **170** (e.g., a pressure surface, a suction surface, a rotor disk

surface, or the like). In various embodiments, the repair blend profile is recessed from the outer surface **164** of the repaired IBR **170**. In various embodiments, the repair blend profile **162** comprises a depth **D1**, a length **L1** and a width **W1**. In various embodiments, the repair blend has a length **L1** that is greater than a width **W1**. However, the present disclosure is not limited in this regard. For example, the length **L1** can be equal to the width **W1**, in accordance with various embodiments. An aspect ratio of a blend profile, as referred to herein, refers to a length **L1** divided by a width **W1** of the repair blend profile. In various embodiments, the repair blend profile is substantially symmetric about a plane defined by a first point, a second point and a third point. The first point can be a max depth location. The second point and the third point can define a line that measures a maximum length (e.g., length **L1**) of the repair blend, in various embodiments. In various embodiments, the second point and the third point can define a line that measures a width in a perpendicular direction from the length **L1**. The present disclosure is not limited in this regard. As referred to herein, “substantially symmetrical” is a first profile on a first side of the plane that is within a profile of between 0.01 inches (0.025 cm) and 0.25 inches (0.64 cm) from the second profile on the second side of the plane, or between 0.01 inches (0.025 cm) and 0.125 inches (0.32 cm), or between 0.01 inches (0.25 cm) and 0.0625 inches (0.16 cm).

(42) Although described herein as being substantially symmetrical, the present disclosure is not limited in this regard. For example, the repair blend profile can comprise a tear drop shape (e.g., repair blend profile **166** from FIG. 5B), a chord reduction (i.e., shorting a chord length by a chord reduction length **C1** along a span of the blade **103** of the repaired IBR **170** as shown in FIG. 5D), or the like. The present disclosure is not limited in this regard.

(43) Referring now to FIG. 6A, a process **200** for repairing an IBR **100** from FIG. 1B from a compressor section (e.g., compressor section **24**) of a gas turbine engine **20** from FIG. 1A is illustrated, in accordance with various embodiments. For example, after a predetermined number of flight cycles, or due to an unscheduled maintenance, the process **200** may be performed for one or more of IBR **100** in the compressor section **24** of the gas turbine engine **20**. In various embodiments, process **200** accumulates data over time, creating greater fidelity and improving future IBR repair determinations, in accordance with various embodiments.

(44) The process **200** comprises inspection of a bladed rotor (e.g., an IBR) (step **202**). In various embodiments, the dimensional inspection can be performed by an inspection system (e.g., inspection system **285** shown in FIGS. 7 and 8 further herein) or visually (e.g., via an inspector or the like). The present disclosure is not limited in this regard. In various embodiments, the dimensional inspection can be facilitated by use of an optical scanner (e.g., structured light scanners, such as white light scanners, structured blue light scanners, or the like) and/or a coordinate-measuring machine. The present disclosure is not limited in this regard. Thus, step **202** can further comprise scanning the IBR **100** with a scanner (e.g., structured light scanners, such as white light scanners, structured blue light scanners, and/or, a coordinate-measuring machine). In response to scanning the IBR **100**, a digital representation of the IBR **100** (e.g., a point cloud, a surface model, or the like) from FIGS. 2-3 is generated. In this regard, the defects **140** of the IBR **100** can be graphically depicted and analyzed, as described further herein. In various embodiments, is either converted to, or is generated as, a three-dimensional model (e.g., a Computer Aided Design (CAD) model or Finite Element Model (FEM)). The three-dimensional model may be utilized for analyzing the IBR **100** in step **204** of process **200**.

(45) In various embodiments, the three-dimensional model generated from step **202** includes a digital representation each defect of the IBR **100** from FIGS. 2-3. In this regard, the process **200** further comprises determining whether the IBR **100** meets serviceable limits (step **204**). In various embodiments, “serviceable limits” as described herein refers to objective parameter limits of defects (e.g., wear, damage, etc.) for the IBR **100**. For example, an objective parameter limit can be a threshold defect depth, a threshold defect length, a threshold defect aspect ratio, a threshold number of defects per blade, a threshold number of defects per IBR **100**, any combination of the

objective parameter limits disclosed herein, or the like. The present disclosure is not limited in this regard, and one skilled in the art may recognize various objective parameters for quantifying whether an IBR meets serviceable limits and be within the scope of this disclosure. In various embodiments, step **204** can be performed by an inspection system **285** or an analysis system **600** as described further herein. The present disclosure is not limited in this regard. If step **204** is performed by the inspection system **285**, a determination on whether the IBR **100** is serviceable can be made rather quickly, on-site at an overhaul facility, in various embodiments. If step **204** is performed by the analysis system **600**, a quick turnaround could still be achieved via cloud computing, or the like, as described further herein, in accordance with various embodiments. In this regard, step **204** can potentially be determined more quickly due to greater potential computing power in a cloud based environment.

(46) In various embodiments, in response to the IBR **100** meeting serviceable limits in step **204**, the process further comprises returning the IBR **100** to service (step **206**). In this regard, the IBR **100** can be re-installed on a gas turbine engine **20** from FIG. **1A**. In various embodiments, the IBR **100** is installed on the same gas turbine engine **20** from which it was removed from. However, the present disclosure is not limited in this regard, and the IBR **100** could be installed on a different gas turbine engine **20** from which it was removed and still be within the scope of this disclosure.

(47) In various embodiments, in response to the IBR **100** not meeting serviceable limits in step **204**, an initial blend profile (e.g., repair blend profile **162**, repair blend profile, **166**, or repair blend profile **168** from FIGS. **5A-D**) is determined and/or generated for each defect **140** of the IBR **100**. In various embodiments, initial blend profiles can only be determined/generated for defects **140** that do not meet the objective defect parameter(s) from step **204**. In various embodiments, a blend profile is generated for all defects **140** of the IBR **100**. The present disclosure is not limited in this regard.

(48) In various embodiments, the blend profile is based on consistent parameters and is generated automatically via the process **200**. For example, each blend may be modeled in step **208** to remove a predetermined amount of additional material and to generate a smooth blend across the entirety of the defect. In various embodiments, the model generated in step **208** is a three-dimensional model (e.g., a Computer Aided Design (CAD) model or Finite Element Model (FEM)). In various embodiments, the model generated in step **208** can be imported into various third party software programs, such as ANSYS, ANSYS Workbench, ANSYS Computational Fluid Dynamics (CFD), or the like. In this regard, structural and aerodynamic simulations can be performed in step **210** for a digital representation of a repaired IBR **170** (a “repaired IBR digital model” as described further herein). The repaired IBR digital model can be a CAD model or an FEM model based on a specific analytical simulation being performed, in accordance with various embodiments. A repaired IBR as described herein refers to a repaired IBR **170** at least one repaired blade portion **172** having at least one repair blend profile **160** (e.g., repair blend profile **162**, **166**, or **168** from FIGS. **5A-D**). The digital representation is based on the inspected IBR that was inspected in step **202** and modeled with initial blend profiles for a plurality of defects **140** in the IBR **100**, in accordance with various embodiments.

(49) In various embodiments, the process **200** further comprises analyzing the repaired IBR digital model (step **210**). In various embodiments, the analysis in step **210** includes various simulations. For example, the various simulations can include simulations to evaluate a modal assurance criteria (MAC), a resonant frequency, an aerodynamic efficiency, a stall margin, damage tolerance, dynamic stress from vibration, or the like. In various embodiments, damage tolerance may be an optional criteria to analyze (e.g., if the bladed rotor is not an IBR but a bladed rotor having a distinct rotor disk and distinct blades). The simulations can be via various simulation software platforms described previously herein.

(50) In various embodiments, analyzing the repaired IBR digital model can include optimizing the blend profile shape based on various parameters. For example, step **210** can include iterating the

blend profile shape based on the various simulations and results of the various simulations. In various embodiments, the blend profile shape is only analyzed for the initial blend profile generated in step **208**. However, the present disclosure is not limited in this regard.

(51) In various embodiment, step **210** further comprises comparing the simulation results to experience based criteria.

(52) For example, if an IBR **100** is known, based on prior test data during development, to have a resonant frequency that varies by a threshold percentage (e.g., 3%, 5%, 8%, etc.) from a nominal frequency, and if a threshold number of the IBR **100** have been in service for a threshold number of cycles (e.g., 2,000 flight cycles, 10,000 flight cycles, or the like), then an experience based criteria for the resonant frequency can be created. For example, if a modal analysis determines a resonant frequency for the repaired IBR digital model is within the threshold percentage, the experience based criteria for resonant frequency would be considered met because the experimental data from development that indicates that the IBR **100** varies in resonant frequency by the threshold percentage without issue has been validated by experience by the threshold number of the IBR **100** having been in service for the threshold number of cycles. Stated another way, the resonant frequency being acceptable within a threshold range of frequencies has been validated by experience in service.

(53) Similarly, if an IBR **100** is known, based on prior test data during development, to have a MAC that varies by a threshold percentage (e.g., 3%, 5%, 8%, etc.) from a nominal MAC, and if the threshold number of the IBR **100** have each been in service for the threshold number of cycles, then an experience based-criteria for the MAC can be created. For example, if the simulation to determine the MAC for the repaired IBR digital model is less than the nominal MAC plus the threshold percentage, the MAC can be determined acceptable as being validated by experience in a similar manner as the resonant frequency example provided above.

(54) If an IBR **100** is known, based on prior test data during development, to have an aerodynamic efficiency that varies by a threshold percentage (e.g., 3%, 5%, 8%, etc.) from a nominal aerodynamic efficiency, and if the threshold number of the IBR **100** have each been in service for the threshold number of cycles, then an experience based-criteria for the aerodynamic efficiency can be created. For example, if the simulation for the repaired IBR digital model has a greater aerodynamic efficiency than the nominal aerodynamic efficiency less the threshold percentage, then the aerodynamic efficiency being acceptable has been validated by experience in the field.

(55) In various embodiments, the acceptable experience based criteria for aerodynamic efficiency can be based on being greater than the aerodynamic efficiency determined from testing and validated by experience in service, as opposed to being based on nominal less the threshold percentage. In this regard, additional margin of error would be provided for the repaired IBR (i.e., a greater safety factor), in accordance with various embodiments. The present disclosure is not limited in this regard.

(56) If an IBR **100** is known, based on prior test data during development, to have a stall margin that varies by a threshold percentage (e.g., 3%, 5%, 8%, etc.) from a nominal stall margin, and if the threshold number of the IBR **100** have each been in service for the threshold number of cycles, then an experience based-criteria for the stall margin can be created. For example, if the simulation to determine stall margin for the repaired IBR digital model has a greater stall margin than the nominal aerodynamic stall margin less the threshold percentage, then the stall margin being acceptable has been validated by experience in the field.

(57) In various embodiments, the acceptable experience based criteria for stall margin can be based on being greater than the nominal stall margin determined from testing and validated by experience in service, as opposed to being based on nominal less the threshold percentage. In this regard, additional margin of error would be provided for the repaired IBR (i.e., a greater safety factor), in accordance with various embodiments. The present disclosure is not limited in this regard.

(58) If an IBR **100** is known, based on prior test data during development, to have a damage

tolerance (e.g., a threshold crack size that results in growth during operation) that varies by a threshold percentage (e.g., 3%, 5%, 8%, etc.) from a nominal damage tolerance, and if the threshold number of the IBR **100** have each been in service for the threshold number of cycles, then an experience based-criteria for the damage tolerance can be created. For example, if the simulation to determine damage tolerance for the repaired IBR digital model has a lower threshold crack growth size than the nominal threshold crack growth size plus the threshold percentage, then the damage tolerance being acceptable has been validated by experience in the field.

(59) Thus, after analyzing the repaired IBR digital model in step **210**, the process **200** further comprises determining whether the experienced based criteria is met for each experience-based criteria parameter (step **212**). In various embodiments, the experience based criteria described herein is exemplary, and not all experience based parameters may be analyzed for a respective repaired IBR.

(60) In various embodiments, in response to the simulation data from the various simulations showing that the experience based criteria is met, the process further comprises performing the repair on the IBR **100** that was inspected in step **202** with a repair process that generates the initial blend profiles (e.g., repair blend profile **162**, **166**, or **168** from FIGS. 5A-D, or the like) for each defect (e.g., defects **140** from FIG. 3) (step **214**). In various embodiments, the repair process may be in accordance with the repair processes described with respect to FIGS. 2-5. In various embodiments, the repair process is performed by a repair system as described further herein. The present disclosure is not limited in this regard.

(61) In various embodiments, with brief reference to FIG. 6B, the process **200** can further comprises vibration testing (step **216**) and correlating the results from the simulations in step **210** (step **218**). Although illustrated as including vibration testing and correlation of results in steps **214**, **216**, the process **200** may disregard these steps and move directly to returning the repaired IBR to service in step **206**, as shown in FIG. 6A. The present disclosure is not limited in this regard. In various embodiments, the vibration testing step **216** can be a bench test (i.e., on a machine configured to induce a response at a specific resonant frequency) or an on-engine test. The present disclosure is not limited in this regard. In various embodiments, results may be correlated based on an analysis of strain gauge data.

(62) Referring back to FIG. 6A, the process **200** further comprises expansion of experience based criteria in step **220**. In this regard, based on additional testing after performing the repair, threshold values can be updated for the experience based criteria to provide greater fidelity and generate faster dispositions for future repair processes, in accordance with various embodiments.

(63) In various embodiments, in response to the experience based criteria not being met, additional analysis of the repaired IBR digital model can be performed in steps **222**, **224**, **226**, and **228**. In this regard, aerodynamic models with greater fidelity can be generated for the CFD analysis in step **222**, and finite element models with greater fidelity (e.g., greater number of nodes or the like) can be generated for the forced response analysis and the structures assessment in step **228**.

(64) In various embodiments, the analysis performed in steps **222**, **224**, **226**, and **228** are performed at a higher fidelity because they will be analyzed as if the repaired IBR is a completely new design. In this regard, the aerodynamic assessment in step **224** and the structures assessment in step **228** can be compared to deterministic criteria (i.e., design criteria or the like). For example, although a resonant frequency of the IBR may not be at a resonant frequency within the experience based range, the resonant frequency may result in a dynamic stress in stress limiting locations that is less than the dynamic stress induced by the IBR **100**, which would result in an IBR **100** that is more robust with regards to dynamic stress relative to an ideal IBR (e.g., a newly manufactured IBR **100**). In various embodiments, even if a resonant frequency that is not with the experience based range of resonant frequencies results in a greater dynamic stress in limiting locations, the dynamic stress can still be less than a threshold dynamic stress based on a Goodman diagram threshold. In various embodiments, the threshold dynamic stress can correspond to a maximum alternating stress

for the material for a mean stress at the location of the dynamic stress. Thus, even if the resonant frequency is outside a range of resonant frequencies that are known to be acceptable based on experience, and the resonant frequency generates a greater dynamic stress relative to the ideal IBR **100**, the repaired IBR **170** can still be acceptable based on the deterministic criteria, in accordance with various embodiments. In this regard, a greater number of acceptable repairs for IBRs can be generated and a greater fidelity can be created for the experience based criteria as the process **200** is repeated for numerous IBRs **100** over time.

(65) In various embodiments, in response to the deterministic criteria being met in steps **230**, subsequent steps **214**, **216**, **218**, **220**, and **206** mentioned previously herein can be repeated. In various embodiments, the criteria expansion of step **220** after going through steps **222**, **224**, **226**, **228**, **230** can greatly widen acceptable experience based criteria as more data of various repair shapes and sizes can be gathered over time.

(66) In various embodiments, in response to the deterministic criteria not being met, the process **200** ends at step **232**. In various embodiments, in response to the process ending at step **232**, the repaired IBR digital model can be stored in a database for later analysis, and the IBR **100** can be placed in storage. In this regard, the repaired IBR digital model could be utilized in later higher fidelity simulations analyzing the IBRs at a system level to determine if the IBR would be acceptable at a system level. Thus, the process **200** can further accommodate system level analysis of IBRs that don't meet the deterministic criteria in step **230**, in accordance with various embodiments.

(67) Referring now to FIG. **6C**, a detailed view of steps **210**, **212** from process **200** are illustrated in accordance with various embodiments as a sub-process (e.g., process **200**). Although illustrated and described herein as a subprocess of the process **200**, present disclosure is not limited in this regard. For example, the process **250** can be utilized independently from the process **200** from FIG. **6A**. Similarly, the process **250** is not meant to limit variations to the process **200** in steps **208**, **210**, **212** as described previously herein.

(68) In various embodiments, the process **250** comprises receiving load inputs for the various simulations described previously herein with respect to the analyzing step **210** from process **200**. For example, the load inputs may include blade definition (e.g., determined from the inspection step **202** from process **200**), blend definition (i.e., determined from step **208** from process **200**), experience based criteria as described previously herein, high cycle fatigue conditions (i.e., boundary conditions for modal analysis), high cycle fatigue capability (e.g., material data described further herein), boundary conditions for fine particle modeling (FPM), a baseline FEM (e.g., a nominal finite element model of an ideal IBR **100**), section files defining blade definition (i.e., two-dimensional cross-sectional contours of a blade determined from a digital point cloud received from the inspection step **202** from process **200**), section files to FEM alignment parameters, aerodynamic analysis boundary conditions, structural analysis boundary conditions, etc. The present disclosure is not limited in this regard. In various embodiments, once the load inputs **252** are uploaded to an analysis system, the process **250** can run upon receiving data from the inspection step **202** as described further herein.

(69) The process **250** further comprises modifying an ideal finite element model (FEM) corresponding to an ideal IBR based on data received from the inspection step (e.g., based on the digital point cloud). In this regard, the ideal finite element model can be modified based on measured geometry of a respective blade **103** in step **202** of process **200**, in accordance with various embodiments.

(70) The process **250** further comprises blending the finite element model based on determining various defect locations (e.g., locations of defects **140** of the IBR **100** from FIG. **2**), and modeling the initial blend determined in step **208** of process **200** (step **256**). In this regard, the repair blend profile **160** can be modeled (e.g., a finite element model) for each respective defect based on a size of the defect **140** (e.g., a depth **D1**, a length **L1**, and a width **W1** of the defect **140** or the like) and a

location of the defect **140** from FIG. 2.

(71) In various embodiments, the process **250** further comprises performing pre-stressed modal analysis with pre and post processing operations of the blended finite element model from step **256** (step **258**). In this regard, steady state analysis, modal analysis, and/or any additional post-processing for the finite element analysis (FEA) and/or pre-stressed modal (PSM) can be performed prior to mode shape analysis and resonant frequency comparisons.

(72) In various embodiments, the process **250** further comprises assessing mode shape shift relative to the ideal IBR (step **260**). The mode shift can be assessed using mode match results. Magnitude of a respective mode shape can be compared to experience based criteria for MAC as described previously herein. In various embodiments, if the MAC does not meet the experience based criteria in step **260**, the process **250** reverts to step **222** of process **200**. In response to the experience criteria for MAC being met, the process **250** assesses frequency shift relative to experience based data as described previously herein (i.e., as described with respect to step **208** of process **200** above) (step **262**). In response to the experience based criteria for frequency shift not being met, the process **250** reverts to step **222** of process **200** as described previously herein. In response to the experience based criteria for frequency shift being met, the process **250** proceeds to a tuned vibratory stress analysis in step **264** and a mistuned vibratory stress analysis in step **266**.

(73) In various embodiments, the tuned vibratory stress analysis is configured to account for any change in mode shape caused by geometric changes from the blend and specific geometry of adjacent blades to a blade being blended as described previously herein. In various embodiments, a tuned stress factor is calculated and applied to the vibratory stress prior to high cycle fatigue analysis. In various embodiments, the tuned vibratory stress analysis performs the comparison by pulling in the FEA and PSM analysis of the baseline from step **268** (i.e., an ideal IBR), in accordance with various embodiments.

(74) In various embodiments, the mistuned vibratory stress analysis in step **266** is configured to account for frequency distribution for all the blades of the inspected IBR **100** from step **202** of process **200**. In this regard, a mistuned stress correction factor can be calculated to adjust for mistuning effects prior to performing a high cycle fatigue assessment in step **276** as described further herein.

(75) After the tuned vibratory stress analysis (step **264**) and the mistuned vibratory stress analysis (step **266**) are performed, a structures assessment, including the high cycle fatigue (HCF) assessment of the IBR **100** with potential repair blend profiles **160**, can be analyzed (step **276**). In various embodiments, the structures assessment performed in step **276** can also include a crack growth assessment (i.e., damage tolerance assessment) to determine a maximum flaw size that the IBR **100** with the potential repair blend profiles **160** can tolerate without growth.

(76) In various embodiments, steps **264**, **266**, **276** are not limited to vibratory stress analysis, mistuning stress analysis, HCF assessment, or crack growth assessment as described previously herein. For example, flutter can also be analyzed by various simulations and be within the scope of this disclosure. Similarly, the analysis can be strictly physics based and/or scaled based on test data, in accordance with various embodiments. In various embodiments, structural simulations for unsteady CFD can also be performed and analyzed and be within the scope of this disclosure.

(77) In various embodiments, the process **250** can extend directly from step **262** to step **212**. For example, the structural simulations in steps **264**, **266**, and the structural assessment in step **276** can be optional, in accordance with various embodiments. For example, if the initial blend definition (e.g., from step **256** or from step **208** in process **200**) is below a threshold blend (e.g., a threshold depth, a threshold aspect ratio, etc.), then steps **264**, **266**, **276** can be skipped, in accordance with various embodiments.

(78) In various embodiments, while the structures analysis is being performed in steps **254**, **256**, **258**, **260**, **262**, **264**, **266**, **268**, a computational fluid dynamics (CFD) analysis can be performed simultaneously in steps **270**, **272**, **274**. For example, in step **270** a cold to hot analysis of the IBR **10**

can be performed prior to performing an aerodynamic analysis. A cold to hot analysis includes the change in the shape of IBR **100** from the conditions the airfoil was inspected at to the engine operating conditions of interest in the aerodynamic analysis based on structural boundary conditions.

(79) In various embodiments, the aerodynamic model can be modified in step **272** to replicate the potential repair blend profile **160** and be as similar as possible to the potential repair blend profile model in step **256** (step **272**). In various embodiments, the aerodynamic model with the potential repair blend profile modeled and the cold to hot analysis serving as inputs can be analyzed via computational fluid dynamics (e.g., via a fine particle model (FPM)) using aerodynamic boundary conditions as described previously herein.

(80) In various embodiments, results from the HCF assessment and the crack growth assessment in step **276**, and results from the CFD analysis from step **274** can be compared to experience based criteria in step **212** of process **200** as described previously herein.

(81) Thus, if the HCF assessment and the crack growth assessment meet the experience based criteria in step **212**, the process proceeds to step **214** of process **200**. If the results from the HCF assessment and the crack growth assessment do not meet the experience based criteria in step **212**, then the process proceeds to step **222** in process **200** as described previously herein.

(82) Referring now to FIG. **6D**, a process **207** is illustrated, in accordance with various embodiments. Although illustrated as having overlapping steps with process **200** from FIG. **6A**, the process **207** can be performed independent of process **200**, in accordance with various embodiments. The present disclosure is not limited in this regard. In various embodiments, the inspection in step **202** can further comprise a visual inspection (step **201**) and a dimensional inspection (step **203**) (e.g., via an inspection system **285** as described further herein). In various embodiments, the visual inspection of step **201** can inform the dimensional inspection of step **203**. For example, areas of interest (e.g., potential defects) can be identified from the visual inspection, and the dimensional inspection in step **203** can be limited to those areas of interest for process **200**, in accordance with various embodiments.

(83) In various embodiments, after the inspection step **202**, if no defects are found in step **205**, the bladed rotor **100** can be returned to service in step **206** as described previously herein. In response to defects being found in step **205**, the defects can be analyzed to determine if the defects are within serviceable limits (step **204**), as described previously herein.

(84) In various embodiments, based on a defect size and/or a defect shape, it can be determined that the defect of the bladed rotor is serviceable as is (step **209**), or an experience based blend can be performed in step **211**. For example, defects below a threshold size can be understood based on experience to be serviceable as is or capable of receiving an experience based blend repair (e.g., a blend having a depth that is less than a depth threshold and a shape that corresponds to an experience based shape). An “experience based shape” as referred to herein is a blend shape that corresponds to a size and configuration that has been performed on a bladed rotor above a threshold number of times (e.g., 100, 1,000, or the like).

(85) In various embodiments, after the experience based blend is performed on the defect in step **211**, the bladed rotor can be re-inspected (step **213**). In various embodiments, the re-inspection can be in accordance with the inspection step **202**.

(86) In various embodiments, the process **207** further comprises determining whether the repaired bladed rotor is within serviceable limits in step **215**. For example, for the experience based blend, the blend can have predefined parameters (e.g., a depth within a specified tolerance, an aspect ratio within a specified tolerance, a radius of curvature within a specified tolerance, etc.). In this regard, the experience based blend can be compared to the limits of an ideal experience based blend to determine if the blend meets the blend definition associated with the ideal experience based blend.

(87) In various embodiments, if the experience based blend is within limits in step **215**, the bladed rotor can return to service in step **206** as described previously herein. If the experience based blend

is not within limits, the process **200** can process to step **217** as described further herein.

(88) In various embodiments, the process **200** can utilize a machine learning system **601** (e.g., a deep neural network (DNN), an artificial neural network (ANN), or the like) further expand criteria in step **220** as described previously herein. In this regard, the process steps in the machine learning system **601** can be fully automated, resulting in a robust, self-learning, inspection, analysis, and repair system as described further herein.

(89) In various embodiments, if the defect is not within serviceable limits in step **204**, the process proceeds to step **217**. In various embodiments, step **217** comprises a detailed inspection of the defect. In various embodiments, the detailed inspection is performed via inspection system **285** as described further herein. In this regard, data corresponding to a size or shape of the defect (or the repair blend profile) can be to perform process **250** in step **219**, in accordance with various embodiments. After having the detailed inspection performed, the bladed rotor can be analyzed in accordance with process **250** from FIG. **6C**, in accordance with various embodiments.

(90) In various embodiments, after performing process **250** in step **219**, the process proceeds to step **214** in response to passing the process **250**. In this regard, a blend corresponding to a repair blend profile from the process **250** can be performed. After the defect found in step **205** is repaired in accordance with the repair blend profile determined from the process **250**, a detailed inspection of the repair blend profile can be performed in step **221**. In various embodiments, if the experience based blend from step **211** is determined not to be within experience based limits in step **215**, the bladed rotor **100** with the experience based blend can proceed to step **221**.

(91) After step **221**, the process **250** can be performed in step **223** for the bladed rotor with the blend repair, in accordance with various embodiments. However, the present disclosure is not limited in this regard, and the process **250** from step **219** can be relied on, in accordance with various embodiments.

(92) In various embodiments, vibrational testing of the repaired IBR can be performed in step **216** in parallel with the process **250** from step **223**. In this regard, the results can be correlated in step **218** (i.e., either to the results from process **250** in step **219** or the process **250** in step **223**). The present disclosure is not limited in this regard.

(93) In various embodiments, in step **225** the structural results can be scaled to operating conditions based on engine test data as described further herein. In response to scaling the results from step **218**, experience based structural criteria can be analyzed (e.g., in accordance with step **212** from process **200**). In response to the experience based structural criteria being met, the bladed rotor can be returned to service in step **206**, in accordance with various embodiments. In various embodiments, if the structural criteria is not met, the process can end at step **232** (e.g., to be held for future repair/use or to be scrapped).

(94) In various embodiments, prior to step **206** and coming from step **205**, **209**, or **215**, a vibration test in accordance with step **216** can optionally be performed to provide additional data for the machine learning system **601**, in accordance with various embodiments. In this regard, the machine learning system **601** can be provided with additional data to use in future determinations, in accordance with various embodiments.

(95) Referring now to FIG. **6E**, a process **227** for inspecting, analyzing, and repairing a bladed rotor **100** is illustrated, in accordance with various embodiments. In various embodiments, the process **227** comprises inspection of the bladed rotor **100** (step **202**) and performing steps **205** and **204** from process **200** in FIG. **6C**. In this regard, the defect has been determined not to be within serviceable limits and proceeds to step **208** from process **200** of FIG. **6A**. In step **208**, an initial blend repair profile is created and sent to the machine learning system **601** from FIG. **6C**. In step **229**, the machine learning system can analyze the initial blend from step **208** and determine whether the initial blend meets the expanded experience based criteria from the learning from the process of FIG. **6C**, in accordance with various embodiments.

(96) In various embodiments, vibration testing can optionally be performed in step **216** after the

analysis in step **229** to further correlate data of the machining learning system **601**.

(97) In various embodiments, in response to the analysis not meeting the expended experience based criteria in step **229**, the process can end at step **232** as described previously herein.

(98) In various embodiments, in response to the machine learning system **601** determining that the expended experience based criteria is met, a system (e.g., system **280** or repair system **290**) described further herein, can perform an automated and adaptive blend based on the repair blend profile from step **208**, in accordance with various embodiments. In this regard, repairs of defects in bladed rotors **100** can be determined and implemented significantly faster relative to typical repair and analysis systems and methods, in accordance with various embodiments.

(99) Referring now to FIG. 7, a system **280** for inspecting, analyzing, and repairing an IBR **100** is illustrated, in accordance with various embodiments. In various embodiments, the system **280** includes an inspection system **285**, an analysis system **600**, and a repair system **290**. Although illustrated as separate systems with separate processors (e.g., processors **286**, **602**, **292**), the present disclosure is not limited in this regard. For example, the system **280** may include a single processor, a single memory, and a single user interface and still remain within the scope of this disclosure.

(100) Similarly, although inspection system **285** and repair system **290** are illustrated as separate systems with separate processors, memories and user interfaces, the present disclosure is not limited in this regard. For example, the inspection system **285** and the repair system **290** may be combined into a single system that communicates with the analysis system **600**, in accordance with various embodiments.

(101) In various embodiments, the analysis system **600** may include one or more processors **602**. The analysis system **600** may be configured to process a significant amount of data during an analysis step as described further herein. In this regard, the analysis system **600** may be configured for remote computing (e.g., cloud-based computing), or the like. Thus, a processing time and a volume of data analyzed may be greatly increased relative to typical repair systems, in accordance with various embodiments.

(102) In various embodiments, the inspection system **285**, the analysis system **600**, and the repair system **290** each include a computer system comprising a processor (e.g., processor **286**, processor(s) **602**, and/or processor **292**) and a memory (e.g., memory **287**, memory **604**, memory **294**). The inspection system **285** and the repair system **290** may each comprise a user interface (UI) (e.g., UI **288**, UI **296**). In various embodiments, the inspection system **285** and the repair system **290** may utilize a single user interface to control both systems. The present disclosure is not limited in this regard.

(103) The analysis system **600** may further comprise a database **606**. In various embodiments, the database **606** comprises various stored data for use in the analysis system **600**. The database **606** may include an inspected IBR database (e.g., with data from various prior inspected IBRs), a repair data database (e.g., with data from various prior repairs performed/approved), a load data database (e.g., with engine load data from structural and/or aerodynamic analysis), a test data database (e.g., with engine specific test data used for validation of structural and/or aerodynamic analysis), a design data database (e.g., with design models having nominal dimensions according to a product definition of the IBR **100**), and/or a material data database (e.g., with material for each component utilized in an analysis step), in accordance with various embodiments.

(104) System **280** may be configured for inspecting, analyzing, and repairing an IBR **100**, in accordance with various embodiments. In this regard, a repair process for an IBR **100** may be fully automated, or nearly fully automated, in accordance with various embodiments, as described further herein.

(105) In various embodiments, systems **285**, **600**, **290** may each store a software program configured to perform the methods described herein in a respective memory **287**, **604**, **294** and run the software program using the respective processor **286**, **602**, **292**. The systems **285**, **600**, **290** may include any number of individual processors **286**, **602**, **292** and memories **287**, **604**, **294**. Various

data may be communicated between the systems **285, 600, 290** and a user via the user interfaces (e.g., UI **288, UI 296**). Such information may also be communicated between the systems **285, 600, 290** and external devices, database **606**, and/or any other computing device connected to the systems **285, 600, 290** (e.g., through any network such as a local area network (LAN), or wide area network (WAN) such as the Internet).

(106) In various embodiments, for systems **285, 600, 290**, each processor **286, 602, 292** may retrieve and executes instructions stored in the respective memory **287, 604, 294** to control the operation of the respective system **285, 600, 290**. Any number and type of processor(s) (e.g., an integrated circuit microprocessor, microcontroller, and/or digital signal processor (DSP)), can be used in conjunction with the various embodiments. The processor **286, 602, 292** may include, and/or operate in conjunction with, any other suitable components and features, such as comparators, analog-to-digital converters (ADCs), and/or digital-to-analog converters (DACs). Functionality of various embodiments may also be implemented through various hardware components storing machine-readable instructions, such as application-specific integrated circuits (ASICs), field-programmable gate arrays (FPGAs) and/or complex programmable logic devices (CPLDs).

(107) The memory **287, 604, 294** may include a non-transitory computer-readable medium (such as on a CD-ROM, DVD-ROM, hard drive or FLASH memory) storing computer-readable instructions stored thereon that can be executed by the processor **286, 602, 292** to perform the methods of the present disclosure. The memory **287, 604, 294** may include any combination of different memory storage devices, such as hard drives, random access memory (RAM), read only memory (ROM), FLASH memory, or any other type of volatile and/or nonvolatile memory.

(108) The system **285, 290** may receive and display information via a respective user interface (e.g., UI **288** and/or UI **296**). The user interfaces (e.g., UI **288** and/or UI **296**) include various peripheral output devices (such as monitors and printers), as well as any suitable input or control devices (such as a mouse and keyboard) to allow users to control and interact with the software program. Although illustrated as not including a user interface, the analysis system **600** is not limited in this regard. For example, a user interface for the analysis system **600** can have a user interface to facilitate load inputs, retrieval of information from the inspection system **285**, or the like.

(109) In various embodiments, inspection system **285** and repair system **290** may each be in electronic communication with analysis system **600**, directly or via a respective user interface (e.g., UI **288** and/or UI **296**). inspection system **285** and repair system **290** may comprise any suitable hardware, software, and/or database components capable of sending, receiving, and storing data. For example, inspection system **285** and/or repair system **290** may comprise a personal computer, personal digital assistant, cellular phone, smartphone (e.g., those running UNIX-based and/or Linux-based operating systems such as IPHONE®, ANDROID®, and/or the like), IoT device, kiosk, and/or the like. Inspection system **285** and/or repair system **290** may comprise an operating system, such as, for example, a WINDOWS® mobile operating system, an ANDROID® operating system, APPLE® IOS®, a LINUX® operating system, and the like. Inspection system **285** and/or repair system **290** may also comprise software components installed on inspection system **285** and/or repair system **290** and configured to enable access to various system **280** components. For example, inspection system **285** and/or repair system **290** may comprise a web browser (e.g., MICROSOFT INTERNET EXPLORER®, GOOGLE CHROME®, APPLE SAFARI® etc.), an application, a micro-app or mobile application, or the like, configured to allow the inspection system **285** and/or repair system **290** to access and interact with analysis system **600** (e.g., directly or via a respective UI, as discussed further herein).

(110) Referring now to FIGS. **8** and **9**, a perspective view of a system **285** for use in an inspection step **202** and/or a repair step **214** of process **200** (FIG. **8**) and a control system **400** for the inspection system **285** (FIG. **9**) are illustrated in accordance with various embodiments. In various

embodiments, the system **285** comprises a repair system **290** and an inspection system **285**. For example, various components of the system **285** may be configured to inspect the IBR **100** and generate a digital map of the IBR **100** (e.g., a point cloud), the inspection system **285** may be configured to transmit the digital map to an analysis system (e.g., analysis system **600**), the inspection system **285** may then receive the results from the analysis system **600**, and/or perform a repair based on a determination from the analysis system **600**, in accordance with various embodiments.

(111) The inspection system **285** comprises a controller **301**, a support structure **302**, a shaft **308**, and a scanner **310**. In various embodiments, the control system **400** comprises the controller **301**, the scanner **310**, a memory **402**, a motor **404**, a database **406**, and sensor(s) **408**, sensor(s) **410**, and inspection component **412**. In various embodiments, the inspection system **285** comprises a device **305** configured for bladed rotor repair and/or bladed rotor inspection.

(112) In various embodiments, the support structure **302** comprises a base **303**, a first vertical support **304**, a second vertical support **306**. In various embodiments, the base **303** may be annular in shape. Although illustrated as being annular, the present disclosure is not limited in this regard. For example, the base **303** may be semi-annular in shape, a flat plate, or the like. In various embodiments, the vertical supports **304**, **306** extend vertically upward from the base **303** on opposite sides of the base (e.g., 180 degrees apart, or opposite sides if the base **303** were a square plate). The shaft **308** extends from the first vertical support **304** to the second vertical support **306**. The shaft **308** may be rotatably coupled to the motor **404**, which may be disposed within the first vertical support **304**, in accordance with various embodiments. The shaft **308** may be restrained vertically and horizontally at the second vertical support **306** but free to rotate relative to the second vertical support about a central longitudinal axis of the shaft **308**. In various embodiments, a bearing assembly may be coupled to the second vertical support **306** to facilitate rotation of the shaft, in accordance with various embodiments.

(113) In various embodiments, the IBR **100** to be inspected in accordance with the inspection step **202** of the process **200** via the inspection system **285** may be coupled to the shaft **308** (e.g., via a rigid coupling, or the like). The present disclosure is not limited in this regard, and the shaft **308** may be coupled to the IBR **100** to be inspected by any method known in the art and be within the scope of this disclosure.

(114) In various embodiments, the scanner **310** is operably coupled to a track system **312**. In various embodiments, the track system **312** may comprise a curved track **314** and a vertical track **316**. The vertical track **316** may slidably couple to the vertical track **316** (e.g., via rollers or the like). The scanner **310** may be slidably coupled to the vertical track **316** (e.g., via a conveyor belt, linkages, or the like). In various embodiments, the scanner **310** is configured to extend from the track system **312** towards the IBR **100** during inspection of the IBR **100** in accordance with step **202** of the process **200**. In this regard, the inspection system **285** may further comprise a control arm **322** (e.g., a robot arm), an actuator (e.g., in combination with the track system **312**) or the like. Although described herein with tracks **314**, **316**, a control arm **322**, and/or an actuator of track system **312**, the present disclosure is not limited in this regard. For example, any electronically controlled (e.g., wireless or wired) component configured to move the scanner **310**, a machining tool (e.g., a mill, a cutter, a lathe, etc.), or the like in six degrees of freedom relative to the IBR **100** is within the scope of this disclosure.

(115) In various embodiments, the inspection component **412** comprises rollers for the curved track, a conveyor belt for the vertical track, and/or a robotic arm coupled to the scanner **310**. In various embodiments, the inspection component **412** comprises only a control arm **322** (e.g., a robot arm). In various embodiments, the inspection component **412** comprises only the rollers for the curved track **314** and the conveyor belt or linkages for the vertical track **316**. The present disclosure is not limited in this regard. In various embodiments, the inspection component **412** is stationary and the IBR **100** being inspected is moveable along three-axis, five-axis, or the like. The

present disclosure is not limited in this regard.

(116) In various embodiments, the scanner **310** comprises a coordinate measuring machine (CMM), a mechanical scanner, a laser scanner, a structured scanner (e.g., a white light scanner, a blue light scanner, etc.), a non-structured optical scanner, a non-visual scanner (e.g., computed tomography), or the like. In various embodiments, the scanner **310** is a blue light scanner. In various embodiments, the scanner **310** may be swapped with another scanner at any point during an inspection step **202** as described further herein. In various embodiments, the inspection system **285** may be configured to swap the scanner **310** with a different scanner during the inspection step **202** of process **200** as described further herein.

(117) A “blue light scanner” as disclosed herein refers to a non-contact structure light scanner. The blue light scanner may have a scan range of between 100×75 mm.sup.2-400×300 mm.sup.2, in accordance with various embodiments. In various embodiments, an accuracy of the blue light scanner may be between 0.005 and 0.015 mm. In various embodiments, the blue light scanner be able to determine distances between adjacent points in the point cloud of between 0.04 and 0.16 mm as measured across three axes. In various embodiments, a volume accuracy of the blue light scanner may be approximately 0.8 mm/m. In various embodiments, a scan depth may be between approximately 100 and 400 mm. In various embodiments, the blue light scanner may comprise a light source including a blue LED. In this regard, the blue light scanner may be configured to emit an average wavelength between 400 and 450 nm, in accordance with various embodiments. Although described with various specifications herein, the blue light scanner is not limited in this regard, and one skilled in the art may recognize the parameters of the blue light scanner may extend outside the exemplary ranges. Use of a blue light scanner provides a high resolution point cloud for a three dimensional object.

(118) In various embodiments, the inspection system **285** further comprises a control arm **320** of the repair system **290**. In various embodiments the control arm **320** comprises a tool holder **321**. The tool holder **321** is configured to couple to a subtractive component **323** (e.g., a mill, a lathe, a cutter, etc.). In various embodiments, the control arm **322** of inspection system **285** may be a control arm for the repair system **290** as well. In various embodiments, the control arms **320**, **322** may be used in both the repair system **290** and the inspection system **285**. The present disclosure is not limited in this regard.

(119) The controller **301** may be integrated into computer system of the inspection system **285** (e.g., in processor **286** and/or memory **287** from FIG. 2B). In various embodiments, the controller **301** may be configured as a central network element or hub to various systems and components of the control system **400**. In various embodiments, controller **301** may comprise a processor (e.g., processor **286**). In various embodiments, controller **301** may be implemented as a single controller (e.g., via a single processor **286** and associated memory **287**). In various embodiments, controller **301** may be implemented as multiple processors (e.g., a main processor and local processors for various components). The controller **301** can be a general purpose processor, a digital signal processor (DSP), an application specific integrated circuit (ASIC), a field programable gate array (FPGA) or other programable logic device, discrete gate or transistor logic, discrete hardware components, or any combination thereof. The controller **301** may comprise a processor configured to implement various logical operations in response to execution of instructions, for example, instructions stored on a non-transitory, tangible, computer-readable medium configured to communicate with the controller **301**.

(120) System program instructions and/or controller instructions may be loaded onto a non-transitory, tangible computer-readable medium having instructions stored thereon that, in response to execution by a controller, cause the controller to perform various operations. The term “non-transitory” is to be understood to remove only propagating transitory signals per se from the claim scope and does not relinquish rights to all standard computer-readable media that are not only propagating transitory signals per se. Stated another way, the meaning of the term “non-transitory

computer-readable medium” and “non-transitory computer-readable storage medium” should be construed to exclude only those types of transitory computer-readable media which were found in *In Re Nuijten* to fall outside the scope of patentable subject matter under 35 U.S.C. § 101.

(121) In various embodiments, the motor **404** of the control system **400** is operably coupled to the shaft **308** of the control system **400**. In various embodiments, the motor **404** may comprise a direct current (DC) stepper, an alternating current (AC) motor or the like. The present disclosure is not limited in this regard. In various embodiments, the sensor(s) **408** include Hall effect sensor(s), optical sensor(s), resolver(s), or the like. In various embodiments, sensor(s) **408** may include sensor(s) configured to detect an angular position of the shaft **308** during an inspection step for an IBR **100** (e.g., step **202** from the process **200**). In this regard, during inspection of the IBR **100**, the controller **301** receives sensor data from the sensor(s) **408**. The controller **301** can utilize the sensor data received from the sensor(s) **408** to correlate an angular position of the IBR **100** being inspected with a location of the scanner **310** as described further herein. In various embodiments, the IBR **100** may remain stationary throughout an inspection process (e.g., inspection step **202** of process **200**) and only a control arm (e.g., control arm **320** and/or control arm **322**) may move. Thus, coordinates of the control arm(s) may be determined via sensor(s) **408** in a similar manner to orient and construct the IBR **100** being inspected.

(122) In various embodiments, the sensor(s) **410** are configured to detect a position of the scanner **310** during the inspection step **202** of process **200**. In this regard, sensor(s) **410** may be position sensors (e.g., capacitive displacement sensors, eddy-current sensors, Hall effect sensors, inductive sensors, optical sensors, linear variable differential transformer (LVDT) sensors, photodiode array sensors, piezoelectric sensors, encoders, potentiometer sensors, ultrasonic sensors or the like). The present disclosure is not limited in this regard. Thus, during inspection of the IBR **100** in accordance with step **202** of process **200**, controller **301** is able to determine a location of the scanner **310** and an angular position of the IBR **100** throughout the inspection. Thus, based on the location of the scanner **310**, an angular location of the IBR **100** and scanning data received from the scanner **310**, a digital map (e.g., a robust point cloud) can be generated during the inspection step **202** of process **200** for the IBR **100** being inspected. In various embodiments, the point cloud encompasses the entire IBR **100** (e.g., between 95% and 100% of a surface area of the IBR **100**, or between 99% and 100% of the surface area of the IBR **100**).

(123) Referring now to FIG. **10**, an analysis system **600** for use in an inspection, analysis, and repair process (e.g., process **200** from FIG. **6A**) as described previously herein is illustrated, in accordance with various embodiments. The analysis system **600** comprises the processor **602**, the memory **604**, and the database **606** from FIG. **7**. In various embodiments, the analysis system **600** is a computer-based system. In various embodiments, the system **600** is a cloud-based computing system. The present disclosure is not limited in this regard. In various embodiments, the analysis system **600** comprises a machine learning system **601** (e.g., a deep neural network (DNN), an artificial neural network (ANN), or the like). In this regard, the analysis system **600** described herein can be configured for machine learning to constantly expand experience based criteria of step **212** in process **200** via step **220** as described previously herein.

(124) The processor **602** is configured to receive an input **603** from the inspection system **285** (e.g., via the control system **400** from FIG. **7**). The input **603** comprises either a point cloud or a three-dimensional model of an IBR **100** inspected in accordance with the process described further herein. In various embodiments, in response to receiving the point cloud, the processor **602** is configured to convert the point cloud to a three-dimensional model (e.g., a finite element model (FEM) for structural analysis, a computation fluid dynamic (CFD) model for aerodynamic analysis, or any other model utilized for analysis).

(125) In various embodiments, the analysis system **600** may include a port configured to couple to a hard drive, or any other device configured to transfer data obtained from inspecting the IBR **100** in step **202** of process **200**. In various embodiments, the processor **602** may be in direct electronic

(e.g., wireless or wired) communication with the inspection system **285** from step **202** of process **200**. In various embodiments, the processor **602** is configured to communicate with the IBR inspection system over a network, or the like. The present disclosure is not limited in this regard. (126) In various embodiments, the processor **602** is in communication with a user interface (“UI”) **630**, which includes a user device **640**. The analysis system **600** can be configured for determining a repaired IBR **170** having repair blend profiles **160** as shown in FIGS. 5A-D, in accordance with various embodiments. In this regard, the processor **602** of the analysis system **600** is configured to receive an input **603** (e.g., from the inspection system **285** of step **202** in process), perform various simulations and analyze the various simulations (e.g., in steps **210**, **222**, **224**, **226**, and/or **228**), and output at the repair blend profiles **160** for respective defects **140** of the IBR **100**. In various embodiments, the analysis system **600** may be configured to perform simulations based on the stack of IBRs **110** from FIG. **1B**. Each IBR in the stack of IBRs **110** can be an IBR **100** that was inspected in step **202** of process **200**. In this regard, a higher fidelity analysis can be performed in steps **222**, **224**, **226**, **228** at a system level for IBR that don't meet the deterministic criteria in step **230** but could meet the deterministic criteria in step **230** based on a modification of an IBR **100** in the stack of IBRs **100**. In this regard, the input **603** may comprise a point cloud or a three-dimensional model for each IBR **100** in a stack of IBRs **110** from FIG. **1B** that have been inspected in accordance with process **500**, in accordance with various embodiments.

(127) In various embodiments, the database **606** includes an inspected IBR database **612** including available IBRs **100** for use in a stack of IBRs **110**. In this regard, the analysis system **600** may be configured to mix and match IBRs **100**, which were on different gas turbine engines **20** from FIG. **1A** previously, based on an optimal repair process, in accordance with various embodiments.

(128) In various embodiments, the analysis system **600** may store a software program configured to perform the methods described herein in the memory **604** and run the software program using the processor **602**. The analysis system **600** may include any number of individual processors **602** and memories **604**. Various data may be communicated between the analysis system **600** and a user via the UI **630** and/or the inspection system **285**. Such information may also be communicated between the analysis system **600** and any other external devices (e.g., a computer numerical control (CNC) machine, an additive manufacturing machine, such as a directed energy deposition (DED) machine, etc.), and/or any other computing device connected to the analysis system **600** (e.g., through any network such as a local area network (LAN), or wide area network (WAN) such as the Internet).

(129) In various embodiments, the processor **602** of the analysis system **600** retrieves and executes instructions stored in the memory **604** to control the operation of the analysis system **600**.

(130) In various embodiments, the database **606** comprises various stored data for use in the analysis system **600** as described further herein. The database **606** may include an inspected IBR database **612**, a repair data database **614**, a load data database **616**, a test data database **618**, a design data database **620**, and/or a material data database **622**, in accordance with various embodiments.

(131) In various embodiments, the inspected IBR database **612** comprises one of a point cloud or a three-dimensional model of inspected IBRs **100** received from the inspection system **285** that are awaiting repair in step **214** of the process **200**. In this regard, the inspected IBR database **612** may include unrepaired IBRs **100** for use in the simulation and analysis steps of process **200** (e.g., steps **210**, **222**, **224**, **226**, and/or **228**). Although described herein as including the inspected IBR database **612**, the present disclosure is not limited in this regard. For example, repair options may be determined for an IBR **100** individually without analysis related to other IBRs **100** in the stack of IBRs **110** from FIG. **1B**, in accordance with various embodiments. Similarly, repair options may be determined for one or more blades **103** of an IBR **100** individually without analysis related to other blades **103** for the IBR **100**, in accordance with various embodiments. However, by increasing the scope to the IBR **100** component level and/or to the rotor module **111** level as described further herein, more optimal repair options may be determined (e.g., based on cost, time, amount of

material removed, etc.), an IBR **100** which may have had a previously unrepairable airfoil determined at a blade level may be repairable based on a component level or module level analysis, and/or blending an airfoil without a defect to offset module or component level effects (e.g., mistuning, aerodynamic capability, etc.), in accordance with various embodiments.

(132) In various embodiments, the repair data database **614** includes previously performed repairs (e.g., blend shapes, additive repair shapes, etc.). In this regard, the repair data database **614** may include any structural debits, aerodynamic debits or the like associated with the previously performed repairs for other IBRs (i.e., not the IBR being inspected). As such, as more repairs are determined, performed, and tested, the repair data database may become more robust, improving the analysis system **600** the more the analysis system **600** is utilized, in accordance with various embodiments.

(133) In various embodiments, the load data database **616** comprises boundary conditions for the gas turbine engine **20** for use in structural analysis and aerodynamic analysis as described further herein. In this regard, for structural analysis, the boundary conditions may include temperature (i.e., highest expected blade temperature, lowest expected blade temperature, etc.), rotor speed (e.g., max rotor speed, typical rotor speed, rotor speed as a function of flight cycle, etc., rotor speed generating modal response, etc.), or any other boundary condition for the IBR **100**, the stack of IBRs **110**, or the high pressure compressor **52**. In various embodiments, module level boundary conditions may include stack stiffness, clocking, clearances (cases, tips, back-bone bending, etc.), blade counts, axial gapping, imbalance, secondary flow influence, or the like.

(134) In various embodiments, the test data database **618** includes engine test data associated with the IBR **100**, the stack of IBRs **110**, and/or the rotor module **111**. For example, prior to certifying a gas turbine engine **20** from FIG. **1A** for production, assumptions with respect to structural analysis performed during a design stage of development may be validated and verified through engine testing. During engine testing, strain gauges may be coupled in various locations on an IBR **100** (e.g., expected high stress locations based on the structural analysis). In response to receiving strain gauge data from the engine testing, analytical or predicted results from the structural analysis can be scaled using actual or measured results to correlate the model to actual data from the engine testing. Thus, the test data database **618** comprises actual test data to be used for scaling predicted data of the analysis system **600** during processes described further herein.

(135) In various embodiments, the design data database **620** comprises three-dimensional models of surrounding components (e.g., blade stages **101**, exit guide vane stage **106**, outer engine case **120**, etc.). In this regard, the analysis system **600** may be configured to prepare a structural model (e.g., via ANSYS, ANSYS Workbench, etc.) and/or a computational fluid dynamics (CFD) model with the surrounding components and the input **603** received from the inspection system **285** and run various simulations with various repair options to determine an optimal repair for an IBR **100**, for each IBR **100** in a stack of IBRs **110**, or for matching repairs of IBRs **100** for various performance parameters (e.g., aerodynamic operability, mistuning, etc.). In various embodiments, the design data database further comprises an original design of the IBR being inspected. In this regard, an original three-dimensional model of the IBR **100** being inspected with nominal dimensions (i.e., nominal in accordance with a product definition of the IBR), in accordance with various embodiments.

(136) In various embodiments, the material data database **622** comprises material data corresponding to a material of the IBR **100**. In various embodiments, the IBR **100** is made of an iron-based alloy (e.g., stainless steel), nickel-based alloy, a titanium alloy, or the like. The present disclosure is not limited in this regard. In various embodiments, material properties for the material the IBR **100** is made of are stored in the material data database **622**. In this regard, in response to performing a structural analysis via the IBR analysis system, the empirical results (after being scaled based on test data from the test data database **618**) may be compared to a threshold zone of acceptance (e.g., a Goodman diagram with steady state stress compared to vibratory stress), where

the threshold zone of acceptance is based on the material properties and a margin of safety, in accordance with various embodiments.

(137) In various embodiments, after the processor **602** performs the various processes disclosed further herein, the processor **602** may output at least one repair process for a respective IBR **100** to the user device (e.g., through the UI **630**, directly to the user device **640**, or the like). In various embodiments, the output may comprise manual instructions for a blend repair process, a computer numerical control (“CNC”) machining process (e.g., blending or the like). In various embodiments, the processor **602** sends CNC machining instructions to the repair system **290** directly, and the repair system **290** repairs the IBR **100** to generate a repaired IBR **170** from FIG. 5A, in accordance with various embodiments.

(138) In various embodiments, the database **606** includes an inspected IBR database **612** including available IBRs **100** for use in a stack of IBRs **110** (i.e., as determined in step **232** of process **200**). In this regard, the analysis system **600** may be configured to mix and match IBRs **100**, which were on different gas turbine engines **20** from FIG. 1A previously, based on IBRs **100** not meeting deterministic criteria at a component level, in accordance with various embodiments.

(139) In various embodiments, the analysis system **600** may store a software program configured to perform the methods described herein in the memory **604** and run the software program using the processor **602**. The analysis system **600** may include any number of individual processors **602** and memories **604**. Various data may be communicated between the analysis system **600** and a user via the UI **630** and/or the inspection system **285**. Such information may also be communicated between the analysis system **600** and any other external devices (e.g., a computer numerical control (“CNC”) machine, etc.), and/or any other computing device connected to the analysis system **600** (e.g., through any network such as a local area network (LAN), or wide area network (WAN) such as the Internet).

(140) In various embodiments, the processor **602** of the analysis system **600** retrieves and executes instructions stored in the memory **604** to control the operation of the analysis system **600**.

(141) In various embodiments, the database **606** comprises various stored data for use in the analysis system **600** as described further herein. The database **606** may include an inspected IBR database **612**, a repair data database **614**, a load data database **616**, a test data database **618**, a design data database **620**, a material data database **622**, and/or a quality data database **624** in accordance with various embodiments.

(142) In various embodiments, the inspected IBR database **612** comprises one of a point cloud or a three-dimensional model of inspected IBRs **100** received from the inspection system **285** that are awaiting repair in step **214** of process **200** or did not meet deterministic criteria in step **230** of process **200**. In this regard, the inspected IBR database **612** may include unrepaired IBRs **100** for use in system level analysis of a stack of IBRs **110**, in accordance with various embodiments. In various embodiments by increasing the scope to the IBR **100** component level and/or to the rotor module **111** level, inspected IBRs that would otherwise be scrapped could potentially be combined in a stack of IBRs to meet satisfy structural criteria and system level aerodynamic criteria, in accordance with various embodiments.

(143) In various embodiments, the repair data database **614** includes previously performed repairs (e.g., blend shapes). In this regard, the repair data database **614** may include any structural debits, aerodynamic debits or the like associated with the previously performed repairs for other IBRs (i.e., not the IBR being inspected). As such, as more repairs are determined, performed, and tested, the repair data database may become more robust, improving the analysis system **600** the more the analysis system **600** is utilized, in accordance with various embodiments.

(144) In various embodiments, the load data database **616** comprises boundary conditions for the gas turbine engine **20** for use in structural analysis and aerodynamic analysis as described further herein. In this regard, for structural analysis, the boundary conditions may include temperature (i.e., highest expected blade temperature, lowest expected blade temperature, etc.), rotor speed (e.g.,

max rotor speed, typical rotor speed, rotor speed as a function of flight cycle, etc., rotor speed generating modal response, etc.), or any other boundary condition for the IBR **100**, the stack of IBRs **110**, or the high pressure compressor **52**. In various embodiments, module level boundary conditions may include stack stiffness, clocking, clearances (cases, tips, back-bone bending, etc.), blade counts, axial gapping, imbalance, secondary flow influence, or the like.

(145) In various embodiments, the test data database **618** includes engine test data associated with the IBR **100**, the stack of IBRs **110**, and/or the rotor module **111**. For example, prior to certifying a gas turbine engine **20** from FIG. 1A for production, assumptions with respect to structural analysis performed during a design stage of development may be validated and verified through engine testing. During engine testing, strain gauges may be coupled in various locations on an IBR **100** (e.g., expected high stress locations based on the structural analysis). In response to receiving strain gauge data from the engine testing, analytical or predicted results from the structural analysis can be scaled using actual or measured results to correlate the model to actual data from the engine testing. Thus, the test data database **618** comprises actual test data to be used for scaling predicted data of the analysis system **600** during processes described further herein.

(146) In various embodiments, the design data database **620** comprises three-dimensional models of surrounding components (e.g., blade stages **101**, exit guide vane stage **106**, outer engine case **120**, etc.). In this regard, the analysis system **600** may be configured to prepare a structural model (e.g., via ANSYS, ANSYS Workbench, etc.) and/or a computational fluid dynamics (CFD) model with the surrounding components and the input **603** received from the inspection system **285** and run various simulations with various repair options to determine an optimal repair for an IBR **100**, for each IBR **100** in a stack of IBRs **110**, or for matching repairs of IBRs **100** for various performance parameters (e.g., aerodynamic operability, mistuning, etc.) (i.e., for component level analysis in steps **222**, **224**, **226** **228**, and/or for system level analysis performed after step **232** in process **200**). In various embodiments, the design data database further comprises an original design of the IBR being inspected. In this regard, an original three-dimensional model of the IBR **100** being inspected with nominal dimensions (i.e., nominal in accordance with a product definition of the IBR), in accordance with various embodiments.

(147) In various embodiments, the material data database **622** comprises material data corresponding to a material of the IBR **100**. In various embodiments, the IBR **100** is made of an iron-based alloy (e.g., stainless steel), nickel-based alloy, a titanium alloy, or the like. The present disclosure is not limited in this regard. In various embodiments, material properties for the material the IBR **100** is made of are stored in the material data database **622**. In this regard, in response to performing a structural analysis via the analysis system **600** (e.g., steps **210**, **222**, **224**), the empirical results (after being scaled based on test data from the test data database **618**) may be compared to a threshold zone of acceptance (e.g., a Goodman diagram with steady state stress compared to vibratory stress), where the threshold zone of acceptance is based on the material properties and a margin of safety, in accordance with various embodiments.

(148) In various embodiments, the quality data database **624** comprises quality data associated with various quality information/dispositions for a known defect shapes, sizes, etc. For example, quality data (e.g., an estimated remaining life, or the like), associated with a known defect can be used for a defect having a size and shape corresponding to a known defect associated with a prior quality determination. In this regard, historical quality data can be utilized by the analysis system **600** for making dispositions and determinations for an inspected IBR as described further herein, in accordance with various embodiments.

(149) In today's digital world, repair operations manuals can be on a computer (i.e., digital); however, engine manuals, serviceable and repair limits, typically remain as hard copies. In various embodiments, based on further defining damage of IBRs via processes **200**, **250** from FIGS. 2A-B, repair limits can be updated digitally as more information is received.

(150) Original Engine Manufacturers (OEMs) typically release a maintenance manual in a hard

copy form to all customers. Each manual can contain a clean, inspection, and/or repair process instructions. The inspection and repair sections also contain documented allowable serviceable and repair size limits. These limits are developed using numerical solutions, tests, or experience.

(151) In response to inspecting damage beyond a limit identified in the maintenance manual, a customer contacts the OEM, and the damage is analyzed by the OEM to determine if the damage is allowable. The new substantiation may use similar methods to the process **250** described herein, and the manual is then changed to include the new limit in response to the new limit being found acceptable. Any new limit request results in additional engine overhaul time turnaround and takes away resources of engineers that could be working on other projects.

(152) In various embodiments, the processes **200**, **250** from FIGS. 2A-B and the analysis system **600** can be utilized to improve serviceable and repair delivery and overall overhaul turnaround time. In this regard, with an extensive database **606**, repair limits can be refined and determined to a greater accuracy and more quickly. In various embodiments, the analysis system **600** can utilize machine learning to further define acceptable serviceable limits for damage. Automated numerical solutions can be used to determine these allowable limits. Cloud services provide easier access to information and direct communication between overhaul facilities and limit substantiation agents. In this regard, a substantiation calculator can be developed to provide acceptable limits as described further herein.

(153) In various embodiments, the inspection system **285** disclosed herein provides greater and more accurate damage dimensions relative to typical systems. Additionally, the inspection system **285** disclosed herein is configured to electronically transfer various details about defects **140** from FIG. 2 as described previously herein, such as shape, size, length, width, depth, etc. In this regard, machine learning algorithms or automated numerical solutions may be accessed from a data depository and a limit determination can be made upon request by a customer as described further herein.

(154) In various embodiments, modifying the manual to just have Clean, Inspect, Repair process and allowing a customer to utilize a “substantiation calculator” can vastly improve overhaul turnaround time. In this regard, a system for accessing the substantiation calculator by the customer can decrease wait times and overall repair cost.

(155) Referring now to FIG. 11, a process **1100** for on-demand analysis of damage to an IBR is illustrated, in accordance with various embodiments. The process **1100** comprises defining damage of an IBR **100**. The damage can be defined by various parameters, such as a maximum length, a maximum width, a maximum depth, or the like. In various embodiments, various additional parameters could be utilized to define the damage and be within the scope of this disclosure. In various embodiments, the damage definition is input into a user interface (e.g., UI **288** of inspection system **285** or the like).

(156) In various embodiments, the process **1100** further comprises comparing the defined damage to dispositioned damage from a disposition database (step **1104**). In various embodiments, the disposition database can be a database in database **606** of the analysis system **600**. In this regard, the UI from step **1102** can be in electronic communication with the analysis system **600** in accordance with any manner previously described herein.

(157) In various embodiments, the process **1100** further comprises determining whether the damage is acceptable (step **1106**), determine whether the damage is acceptable with a repair in response to determining the damage is not acceptable (step **1108**), and determining the bladed rotor should be scrapped in response to determining the damage is not acceptable with a repair (step **1110**).

(158) In various embodiments, the process further comprises generating a report of the disposition back to the user interface utilized in step **1102** (step **1112**). In this regard, based on machine learning algorithms from the data depository (e.g., database **606** of analysis system **600**), damage limit determinations can be made on demand at overhaul facilities, in accordance with various embodiments.

(159) Benefits, other advantages, and solutions to problems have been described herein with regard to specific embodiments. Furthermore, the connecting lines shown in the various figures contained herein are intended to represent exemplary functional relationships and/or physical couplings between the various elements. It should be noted that many alternative or additional functional relationships or physical connections may be present in a practical system. However, the benefits, advantages, solutions to problems, and any elements that may cause any benefit, advantage, or solution to occur or become more pronounced are not to be construed as critical, required, or essential features or elements of the disclosure. The scope of the disclosure is accordingly to be limited by nothing other than the appended claims, in which reference to an element in the singular is not intended to mean “one and only one” unless explicitly so stated, but rather “one or more.” Moreover, where a phrase similar to “at least one of A, B, or C” is used in the claims, it is intended that the phrase be interpreted to mean that A alone may be present in an embodiment, B alone may be present in an embodiment, C alone may be present in an embodiment, or that any combination of the elements A, B and C may be present in a single embodiment; for example, A and B, A and C, B and C, or A and B and C. Different cross-hatching is used throughout the figures to denote different parts but not necessarily to denote the same or different materials.

(160) Systems, methods, and apparatus are provided herein. In the detailed description herein, references to “one embodiment,” “an embodiment,” “various embodiments,” etc., indicate that the embodiment described may include a particular feature, structure, or characteristic, but every embodiment may not necessarily include the particular feature, structure, or characteristic. Moreover, such phrases are not necessarily referring to the same embodiment. Further, when a particular feature, structure, or characteristic is described in connection with an embodiment, it is submitted that it is within the knowledge of one skilled in the art to affect such feature, structure, or characteristic in connection with other embodiments whether or not explicitly described. After reading the description, it will be apparent to one skilled in the relevant art(s) how to implement the disclosure in alternative embodiments.

(161) Numbers, percentages, or other values stated herein are intended to include that value, and also other values that are about or approximately equal to the stated value, as would be appreciated by one of ordinary skill in the art encompassed by various embodiments of the present disclosure. A stated value should therefore be interpreted broadly enough to encompass values that are at least close enough to the stated value to perform a desired function or achieve a desired result. The stated values include at least the variation to be expected in a suitable industrial process, and may include values that are within 10%, within 5%, within 1%, within 0.1%, or within 0.01% of a stated value. Additionally, the terms “substantially,” “about” or “approximately” as used herein represent an amount close to the stated amount that still performs a desired function or achieves a desired result. For example, the term “substantially,” “about” or “approximately” may refer to an amount that is within 10% of, within 5% of, within 1% of, within 0.1% of, and within 0.01% of a stated amount or value.

(162) Furthermore, no element, component, or method step in the present disclosure is intended to be dedicated to the public regardless of whether the element, component, or method step is explicitly recited in the claims. No claim element herein is to be construed under the provisions of 35 U.S.C. 112(f) unless the element is expressly recited using the phrase “means for.” As used herein, the terms “comprises,” “comprising,” or any other variation thereof, are intended to cover a non-exclusive inclusion, such that a process, method, article, or apparatus that comprises a list of elements does not include only those elements but may include other elements not expressly listed or inherent to such process, method, article, or apparatus.

(163) Finally, it should be understood that any of the above described concepts can be used alone or in combination with any or all of the other above described concepts. Although various embodiments have been disclosed and described, one of ordinary skill in this art would recognize that certain modifications would come within the scope of this disclosure. Accordingly, the

description is not intended to be exhaustive or to limit the principles described or illustrated herein to any precise form. Many modifications and variations are possible in light of the above teaching.

Claims

1. A method, comprising: determining, via a processor, whether serviceable limits are met for an inspected bladed rotor; automatically generating, via the processor, a three-dimensional model including a repair blend profile for a defect in response to determining the inspected bladed rotor does not meet the serviceable limits; automatically performing, via the processor, a first set of simulations on the three-dimensional model; automatically determining, via the processor, whether a first set of results of the first set of simulations meet an experience-based criteria; automatically performing, via the processor, a second set of simulations on the three-dimensional model in response to a first result in the first set of results not meeting a first objective parameter in the experience-based criteria; and automatically determining, via the processor, whether a second set of results and the first set of results meet a deterministic criteria.
2. The method of claim 1, further comprising repairing the inspected bladed rotor with the repair blend profile in response to determining the first set of results and the second set of results meet the deterministic criteria.
3. The method of claim 2, further comprising modifying the experience-based criteria in response to determining the first set of results and the second set of results meet the deterministic criteria.
4. The method of claim 1, wherein the experience-based criteria includes a modal assurance criteria (MAC), a resonant frequency, an aerodynamic efficiency, and a stall margin.
5. The method of claim 1, further comprising receiving, via the processor, a digital point cloud of the inspected bladed rotor.
6. The method of claim 5, further comprising automatically converting, via the processor, the digital point cloud to the three-dimensional model with the repair blend profile for the defect in response to determining the inspected bladed rotor does not meet the serviceable limits.
7. The method of claim 1, wherein the serviceable limits include at least one of a threshold defect depth, a threshold defect length, a threshold defect aspect ratio, a threshold number of defects per blade, a threshold number of defects per bladed rotor.
8. A method, comprising: inspecting, via an inspection system, an inspected bladed rotor; analyzing, via an analysis system, the inspected bladed rotor, the analyzing including: determining whether serviceable limits are met for the inspected bladed rotor; determining whether experience-based criteria is met for a repaired bladed rotor in response to the inspected bladed rotor not meeting the serviceable limits, the repaired bladed rotor including a repair blend profile for repairing a defect; and determining whether deterministic criteria is met for the repaired bladed rotor in response to the repaired bladed rotor not meeting the serviceable limits; and repairing, via a repair system, the inspected bladed rotor to form the repaired bladed rotor in response to determining the inspected bladed rotor does not meet the serviceable limits and determining one of experience-based criteria for the repaired bladed rotor being met or the deterministic criteria for the repaired bladed being met.
9. The method of claim 8, wherein the inspecting is performed with a structured scanner to form a point cloud of the inspected bladed rotor.
10. The method of claim 9, wherein the analyzing includes generating a repaired bladed rotor digital model of the repaired bladed rotor based on the point cloud and determining an initial blend shape for the repair blend profile.
11. The method of claim 10, wherein the analyzing further comprises performing a set of simulations on the repaired bladed rotor digital model.
12. The method of claim 11, wherein the analyzing further comprises comparing a set of results from the set of simulations to the experience-based criteria.

13. The method of claim 12, wherein the repairing comprises performing a repair on the inspected bladed rotor to generate the repair blend profile in response to the experience-based criteria being met by the set of results from the set of simulations.
14. The method of claim 12, wherein the experience-based criteria includes a modal assurance criteria (MAC), a resonant frequency, an aerodynamic efficiency, and a stall margin.
15. A system, comprising: an inspection system configured to generate a point cloud of at least a portion of an inspected bladed rotor; and an analysis system in electronic communication with the inspection system, the analysis system comprising a tangible, non-transitory computer-readable storage medium having instructions stored thereon that, in response to execution by a processor, cause the processor to perform operations comprising: receive, via the processor, the point cloud; determine, via the processor and based on the point cloud, whether serviceable limits are met for the inspected bladed rotor; generate, via the processor, a three-dimensional model with a repair blend profile for a defect in the inspected bladed rotor in response to determining the inspected bladed rotor does not meet the serviceable limits; perform, via the processor, a set of simulations of the three-dimensional model; and determine, via the processor, whether an experience-based criteria validated by a threshold number of flight cycles is met by a first set of results of the set of simulations.
16. The system of claim 15, wherein the inspection system comprises a scanner.
17. The system of claim 15, wherein the experience-based criteria includes a modal assurance criteria (MAC), a resonant frequency, an aerodynamic efficiency, and a stall margin.
18. The system of claim 15, wherein the operations further comprise, performing additional simulations, via the processor, in response to determining the first set of results do not meet the experience-based criteria and determining, via the processor, whether a second set of results and the first set of results meet a deterministic criteria, the deterministic criteria having a different range than the experienced-based criteria.
19. The system of claim 15, wherein the operations further comprise, sending repair instructions to the inspection system in response to determining the first set of results meet the experience-based criteria.
20. The system of claim 19, wherein the inspection system comprises a computer numerical control (CNC) machine, the CNC machine configured to perform the repair instructions.
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